

Time: Tuesday and Thursday, 02:00 - 03:15 PM

Location: CAPF 4031

Instructor: Shun Dong

Office: CAPF 3128

Office hours: Tuesday and Thursday, 01:00 - 02:00 PM

Or by appointment

Zoom link: <https://kansas.zoom.us/j/85335100900> Passcode: 914392

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COURSE PREREQUISITES

Corequisites: IST 202 or BSAN 310 and IST 310

COURSE DESCRIPTION

Database management systems are central to modern business information systems. They facilitate the sharing of data across the organization and therefore support the notion that data is a corporate resource. Organizations increasingly store and rely on vast amounts of data related to their various stakeholders, including customers, employees, and suppliers. The ability to sense, harness, retrieve, and provide this data is greatly enhanced by ever-growing storage capacities and computing power. However, firms are simultaneously facing escalating information overload, which threatens to hinder their functionality. Effective data management is therefore crucial for organizational success. Moreover, data synthesis and analysis promise to support business strategy and offer a competitive advantage.

This is a **3-credit-hour**, semester-long course, with two **75-minute in-person sessions** on **Tuesdays and Thursdays from 2:00 PM to 3:15 PM in CAPF 4031**. Students should expect to spend at least two hours per week on additional study and preparation for each credit hour.

The course introduces the fundamental concepts of data management, database systems, and their applications in business. It aims to provide sufficient technical details while focusing on the organizational and implementation challenges of managing data within an organizational context. Key topics include relational data modeling, logical database design, structured query language (SQL), and the role of data in organizations.

LEARNING OUTCOMES

Students will learn about the design, implementation, use, and management of database management systems. The course material is divided into two related components. The first component will focus on data models and logical database design and will equip students to:

- Explain the advantages and disadvantages of the database approach to data management
- Understand the core concepts of database systems and their importance in various applications.
- Describe the components of a relational database system.
- Design conceptual models and relational models for relational databases using proper modeling techniques and syntax.
- Apply the principles of normalization to convert a conceptual design to a database relational model.
- Write SQL queries to create and retrieve data from relational databases.
- Gain exposure to emerging trends in database technologies.
- Learn how to leverage Artificial Intelligence (AI), Cloud Computing, and other technologies for problem-solving using data management and SQL querying.

TEXTS & MATERIALS

The primary reference materials for this course are the slides and articles linked/uploaded on Canvas. ***There is no textbook required for this course.*** Students may, however, wish to supplement the in-class and online material provided with the following textbook:

Hoffer, Jeffrey A., Ramesh, Venkataraman, and Topi, Heikki, Modern Database Management. 12th Edition, Prentice-Hall, 2016) (ISBN-13: 978-126-948-947-8)

CLASS FORMAT

This course is being offered in person. All the learning materials for this course will be accessible via Canvas.

All deliveries (i.e., assignments, project reports, etc.) must be submitted via Canvas. The group assignments and project report must be submitted via Canvas before the respective deadlines.

GRADING POLICY

Your performance in this course will be evaluated in four areas: class participation, two midterm exams, five individual homework assignments, in-class quiz (for participation) and task, and a group project.

Class deliverables and the scale for overall grade are given below:

Deliverable	Percentage of grade
Homework	25%
In-class Tasks and Quizzes	10%
Midterm Exam I	20%
Midterm Exam II	20%
Group Project	25%

Note: Final course grades will be scaled to the above grading categories (e.g., if there are 100 homework points possible these points will be scaled to reflect a “worth” of 25% of the overall grade).

Grade	Percentage
A	92.5–100%
A–	87.5–92.5%
B+	82.5–87.5%
B	77.5–82.5%
B–	72.5–77.5%
C±	60–72.5%
D±	50–59.5%
F	0–49.5%

You may not submit extra work to raise your grade, unless I have specified at the outset of class that such an opportunity exists. I reserve the right to amend this syllabus, assignments, and due dates throughout the semester.

Grade Dispute Policy: If you disagree with the grade, please inform me by writing via email within one week of the day on which I returned your work. The note should include specific details as to what part of the submission was graded incorrectly. If you appeal, I will re-grade your entire exam, quiz, or assignment; your grade may increase, decrease, or remain the same.

IN-CLASS TASKS, QUIZZES & CLASS PARTICIPATION

Across the semester, there will be randomly assigned in-class tasks (e.g., problem-solving exercises, lab questions). These tasks will be completed during class time, and you are required to submit your solutions by the end of that day.

Please note that class participation assesses your overall engagement and involvement in the course, while the in-class tasks specifically reflect the work you complete during class.

Your attendance will be evaluated primarily through randomly assigned in-class quizzes. These quizzes will be relatively straightforward, often consisting of only one or two questions (and in some cases, possibly no written questions at all).

Together, in-class tasks and quizzes will account for 10% of your total course grade.

HOMEWORK ASSIGNMENTS

There will be **five homework assignments** due throughout the semester. The assignments are designed to reinforce key concepts and material presented during lectures and labs and will therefore consist of a variety of both free-thought and task-based questions. The assignments will also aid students in preparing for the two midterms exams, as well as the final project.

EXAMS

Midterm Exam 1 will cover the material discussed during the first part of the course (see the breakdown of course below) and will consist of multiple question formats (i.e., multiple choice, short answer, etc.).

Midterm Exam 2 will consist of traditional questions (multiple choice, etc.) and SQL-based questions, evaluating your ability to write SQL.

Concept	Assignments	Exams
• Database Systems • Database Development Process		Midterm Exam I (Oct. 9th)
• Entity-Relationship Model • Enhanced ER Model	Assignment I	
• Logical Design: Converting ER to Relational Tables		
• Normalization	Assignment II	
Introduction to SQL o Data Control Language o Data Definition Language o Data Manipulation Language o Simple Queries	Assignment III	Midterm Exam II (Nov. 20th)
SQL Programming: o Aggregate Queries o Join Queries	Assignment IV	
Advanced SQL programming: o Subqueries and inline queries	Assignment V	
• Advanced Topics		

Misconduct During Exams: You will have access to your laptop during the exams. However, you may **NOT** use your laptop, your phone, or any other device to communicate with anyone during an exam. If I find that you violated the KU Honor Code, you will receive an “F” for the course, and the Registrar will be notified that the grade was given because of academic misconduct.

Exam Make-up Policy: Alternative midterm times are only possible with **one-week advance notice** and appropriate justification. The midterm must be taken prior to its scheduled time. Once the midterm has been held as scheduled, a make-up midterm is prohibited. Even if the midterm absence is uncontrollable, no make-up midterm is allowed. The dual grading scheme is created to accommodate these unexpected absences. An alternative final exam time must be approved by the Office of the Dean of Students.

GROUP PROJECT

It is highly likely that you will work as a team member in your future vocation, and it is therefore beneficial to learn effective teamwork practices. Also, in your future organization, you will be required to work with peers with complimentary skills for each project, whom you may or may not know beforehand. The group project will provide you with an opportunity to apply everything you learn to develop a deliverable by the end of the semester with formative learning experience.

You will be assigned to a group by the instructor at the outset of the semester. As a group you will choose a “client” (e.g., Netflix, Amazon, etc.) for whom you will develop a database system. The client should have data recorded from multiple sources (e.g., customers, employees, etc.) and a need for this data to be updated on a daily basis (e.g., customer purchases). Upon choosing a client, you will then describe the business use-case, which will entail providing a description of the business and how it will use the database you design in the business’ day-to-day operations. You will then design the database and implement it over the course of the semester. As part of the project, you are *not* expected to collect real-world data, but will instead create fictitious data (e.g., make up customer information, sales records, etc.) and therefore you need not have access to any propriety technology or data. There are two project update documents you need to submit, as specified in the schedule. These are brief summary documents that will nominally count towards the project grade and will serve to keep you on track with the project work.

The culmination of the project will be a functional database. You will make a **15-20 minute group presentation** and **submit an executive report** at the end of the semester as project deliverables to be graded. The final project report should have all the following elements: introduction, problem statement, requirement summary, conceptual model, relational database, normalization, data dictionary, sample data screenshots, implementation details, SQL queries, assumptions, and conclusions. The explicit requirements and grading criteria for the report will be specified in a detailed document uploaded on Canvas.

The group writing assignments will help you improve your writing effectiveness in preparing professional reports. Each member of the group is expected to write a section of the report, based on the work that he/she has done. Prepare written submissions using a word processor, single-spaced, and styled appropriately for business correspondence. For technical assistance with writing, you can take advantage of resources provided by the KU writing center <http://writing.ku.edu/>.

At the end of the course, each of you will get a performance and productivity peer evaluation form which you will submit to the instructor. In this form, you can summarize your contribution as well as the contribution of other members in your group. Peer evaluations *will* be considered when final project grades are determined.

Additional details on the group project will be provided.

CLASS SCHEDULE

A detailed schedule of contents covered in each lecture/session is shared on Canvas. Minor updates to it will be made periodically by the instructor, as the course progresses, and all students will be notified if there are changes to due dates of course deliverables.

COURSE POLICIES

Communication. Students can expect to receive weekly communications from the instructor (via the “Announcements” section on Canvas) and are responsible for the information communicated therein. Students are also responsible for any official correspondence sent to/through their University of Kansas email address. Students are expected to use their University-provided email address to communicate with the instructor and can *expect a response within 24-48 hours*. If a response is not received within 48 hours, Students are encouraged to send a follow-up email (things can, at times, fall through the cracks). Privacy considerations, such as federal law, may apply when using a non-University address for communication.

Timeliness. Late work will be penalized. All the assignments and quizzes are due by 11:59PM on the specified in the course schedule. Late submissions of assignments may attract a penalty. If you have obligations that conflict with exam/assignment/report due dates, you should contact the instructor as soon as possible and ahead of the due date. Only in extremely rare circumstances will make-up exams be permitted after-the-fact. Equipment failure or conflicting time availability of team members is not an acceptable reason for turning in an assignment late. **Late submissions will be penalized 25% of the assignment total each 24-hour period** (i.e., after 72 hours, a late assignment will be assigned a 0).

Score/Grade appeals. If you feel that an assigned grade inaccurately reflects your performance, feel free to discuss the matter with the instructor. Timeliness is the responsibility of the student, however. As such, grades that have been assigned for more than one month are not open for appeal.

Software support. The instructor will provide software support for the software that is explicitly being used for course work. Students will be responsible for troubleshooting virtually all potential software issues encountered when using non-course-based software, including operating system issues.

ACADEMIC CODE OF HONOR

The KU School of Business seeks to develop future leaders with the highest ethical standards. It is through a strong code of conduct that we maintain mutual trust and respect among students, faculty, and staff. This code of conduct was developed by the students, faculty, and staff to articulate the School's core values and provide guidance on academic integrity. This code applies to the conduct of students, faculty, and staff at any function or academic activity conducted by the School of Business at the University of Kansas. See <https://business.ku.edu/services/student-academic-services/honor-code>.

Academic misconduct in this class will not be tolerated. Academic misconduct is not only an unethical behavior; it also deprives you of educational opportunities. There are many forms of cheating, including but not limited to:

- Possession of course material from previous sessions of this course.
- Plagiarism. All written work will be monitored by plagiarism detection software subscribed to by KU.
- Giving, receiving or unauthorized uses of aids during tests or giving test questions to another student who has not yet taken a given test.
- Following KU policy, posting content found on Blackboard/Canvas or any other class materials to a public website is a violation of intellectual property rights.
- Disrupting class is a violation of the honor code and is classified as behavioral misconduct.

ACCOMMODATIONS FOR STUDENTS WITH DISABILITIES

The Student Access Center (SAC) coordinates academic accommodation and services for all eligible KU students with disabilities. If you have a disability for which you wish to request accommodation and have not contacted SAC, please do so as soon as possible. The SAC is located in 22 Strong Hall and can be reached at 785-864-4064 (V/TTY). Information about the SAC's services can be found at <https://access.ku.edu/>. Please also contact the instructor privately in regard to your needs in this course.

POSITIVE CODE OF CONDUCT (R-E-S-P-E-C-T)

The School of Business promotes a Positive Code of Conduct based on seven principles:

Responsibility: We will accept and be accountable for the outcomes of our actions, work to exceed expectations, meet established deadlines for assignments, and represent the School of Business in a positive manner.

Enthusiasm: We will exhibit leadership, drive and passion in all of our pursuits both inside and outside of classroom by actively engaging room discussions, events, and student organizations.

Self-Development: We will challenge ourselves to seek out opportunities for self-development, for cultivating belief in ourselves will enable us to foster respect for others.

Professional Integrity: We will be honest, trustworthy and genuine in all interactions.

Equity: We will contribute to the learning community at least as much as we receive, support equal opportunity for all students to learn, and act with respect toward the community and the environment.

Compassion: We will practice care, empathy, and understanding toward others by providing them with support or feedback when it is needed.

Teamwork: We will create an open atmosphere where everyone can contribute ideas and engage in respectful discussion without undue criticism, recognizing the value of a team in working toward a common goal.

COMMERCIAL NOTETAKING

Pursuant to the University of Kansas' Policy on Commercial Note-Taking Ventures, commercial notetaking is not permitted in this class. Lecture notes and course materials may be taken for personal use, for the purpose of mastering the course material, and may not be sold to any person or entity in any form. Any student engaged in or contributing to the commercial exchange of notes or course materials will be subject to discipline, including academic misconduct charges, in accordance with University policy. Notetaking provided by a student volunteer for a student with a disability, as a reasonable accommodation under the ADA, is not the same as commercial notetaking and is not covered under this policy.

COMMERCIAL NOTE-TAKING

Pursuant to the University of Kansas' Policy on Commercial Note-taking Ventures, commercial notetaking is not permitted in FIN 415, Corporate Finance. Lecture notes and course materials may be taken for personal use and for the purpose of mastering the course material. However, students are not authorized to upload/share/sell class material in any form to any person or entity. Any student engaged in or contributing to the commercial exchange of notes or course materials will be subject to discipline, including academic misconduct charges, in accordance with university policy. Notetaking provided by a student volunteer for a student with a disability, as a reasonable accommodation under the ADA, is not the same as commercial notetaking and is not covered under this policy. Please refer to the following link for additional details: <https://policy.ku.edu/provost/commercial-note-taking>.

EQUAL OPPORTUNITY

The University of Kansas is proud of its goal to help all individuals realize their potential. To this end, the university is committed to providing an equal opportunity for all qualified individuals to be considered for employment, benefits and conditions of employment, educational programs and activities, regardless of race, religion, color, ethnicity, sex, disability, national origin, ancestry, age, status as a veteran, sexual orientation, marital status, parental status, gender identity, gender expression, or genetic information.

DIVERSITY & INCLUSION

As a premier international research university, the University of Kansas is committed to an open, diverse, and inclusive learning and working environment that nurtures the growth and development of all. KU holds steadfast in the belief that an array of values, interests, experiences, and intellectual and cultural viewpoints enrich learning and our workplace. The promotion of and support for a diverse and inclusive community of mutual respect require the engagement of the entire university.

CONCEALED CARRY LAW

Individuals who choose to carry concealed handguns are solely responsible to do so in a safe and secure manner in strict conformity with state and federal laws and KU weapons policy (refer to <http://concealedcarry.ku.edu/>). Safety measures outlined in the KU weapons policy specify that a concealed handgun:

- Must be under the constant control of the carrier.

- Must be out of view, concealed either on the body of the carrier, or backpack, purse, or bag that remains under the carrier's custody and control.
- Must be in a holster that covers the trigger area and secures any external hammer in an un-cocked position.
- Must have safety on and have no round in the chamber.

Instructors are allowed by Kansas Board of Regents policy, to require backpacks, purses and other bags be placed at the front of the classroom during exams and quizzes, and as such those items will not be under the constant control of the individual. Students who choose to carry a concealed handgun in a purse, backpack, or bag must review and plan each day accordingly, and are responsible for making alternate arrangements as necessary. The university does not provide appropriate secured storage for concealed handguns. Individuals who violate the KU weapons policy may be asked to leave campus with the weapon and may face disciplinary action under the appropriate university code of conduct.

FREEDOM OF EXPRESSION

Encouraging students to engage in debate and deliberation on any topic in an effective and responsible manner is an essential part of the educational mission of a university. This course does not focus on sensitive or potentially divisive social issues, but regardless it is not the proper role of a university to insulate individuals from ideas and opinions they might find challenging, unwelcome, disagreeable, or even offensive. All university community members share a responsibility to maintain a climate of mutual respect, but concerns about civility and mutual respect should never be used to justify silencing the discussion of ideas. Therefore, consistent with principles of free expression embraced by Purdue University, the University of Chicago, and other institutions, the instructor supports freedom of expression for students to discuss any course-related topic that presents itself.

ADDITIONAL LINKS

See below for additional information regarding KU and Business School policies and procedures.

- **University Academic Support Centers**
 - <https://access.ku.edu/>
- **The School of Business Positive Code of Conduct**
 - <https://business.ku.edu/honor-code/code-of-conduct>
- **Concealed carry**
 - <http://concealedcarry.ku.edu/>
- **Academic Misconduct**
 - <https://policy.ku.edu/governance/USRR#art2sect7>
- **Change of Grade Required**
 - <https://policy.ku.edu/registrar/grade-change;>
 - <https://policy.ku.edu/governance/USRR#art2sect3>
- **Diversity and Inclusion**
 - <https://policy.ku.edu/provost/diversity-inclusion>
- **Mandatory Reporting (Title IX/Civil Rights)**
 - <https://policy.ku.edu/civil-rights/mandatory-reporting>
- **Nondiscrimination, Equal Opportunity, and Affirmative Action**
 - <https://policy.ku.edu/IOA/nondiscrimination>
- **Sexual Harassment**
 - <https://policy.ku.edu/civil-rights/sexual-harassment>
- **Student Rights and Responsibilities**
 - <https://policy.ku.edu/student-affairs/student-code>
- **Racial and Ethnic Harassment Policy**
 - <https://policy.ku.edu/civil-rights/racial-ethnic-harassment-policy>